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3.2.4 Modified Gram-Schmidt (MGS)

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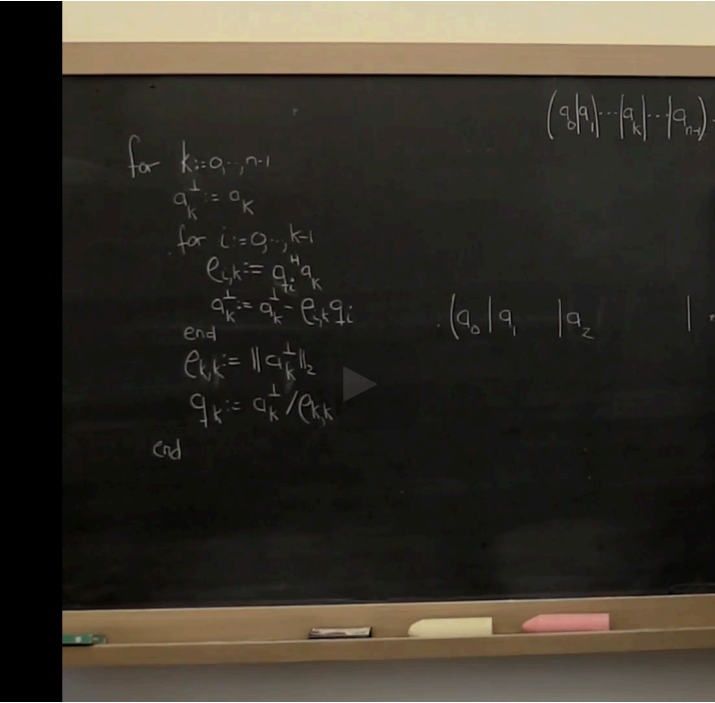
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So but now you understand how the Gram Schmidt Orthogonalization process

works. And what I've done here is I've written it down in great detail. And how

does this work? Well, we march over all of the columns of

A. When we get to a_k, we will assume that we have already computed all of the

previous columns of Q and that we have computed a big part the top left parts

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Reading assignment

1.0/1.0 point (graded)

Read Unit 3.2.4 of the notes. [\[LINK\]](#)

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[Possible administrative error in HW Assignment--wrong MATLAB file](#)

2

Instructions call for implementing the function MGS_QR(A) in the MATLAB fil...

Homework 3.2.4.1

1.0/1.0 point (graded)

the point (grades,
Assume that $q_0, \dots, q_{k-1}$ are mutually orthonormal. Let $\rho_{j,k} = q_j^H y$ for $j = 0, \dots, i-1$. Show that

$$\underbrace{q_i^H y}_{\rho_{i,k}} = q_i^H (y - \rho_{0,k} q_0 - \dots - \rho_{i-1,k} q_{i-1})$$

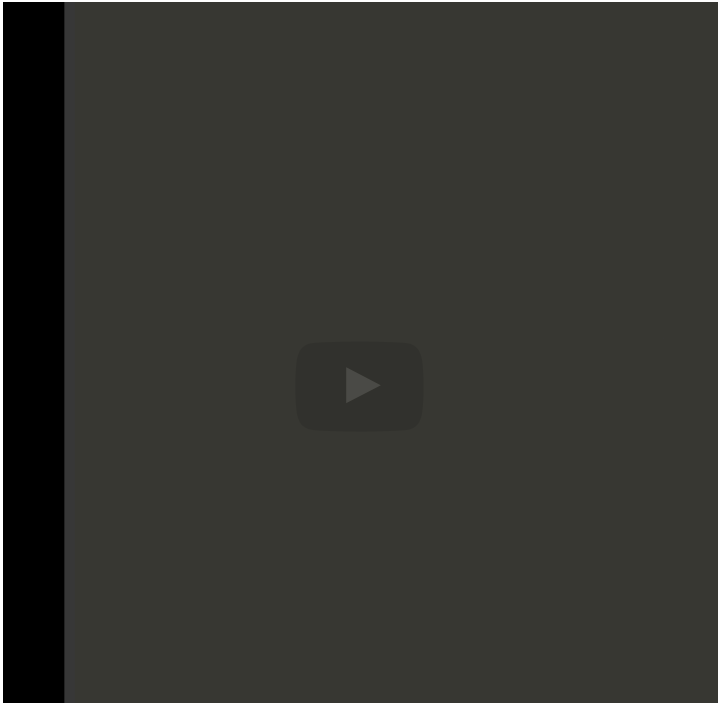
for $i = 0, \dots, k-1$.

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Okay, back to the top of the loop. Now we look at the next column of A. What do we do? In the first iteration of this, we subtract off, we compute rho_02. And we can use that to subtract off the component in the direction of q_0. But notice that in the next iteration then, we don't look at the dot product of q_1 with a_2. We instead look at the dot product of q_1 with this updated vector here. But what you

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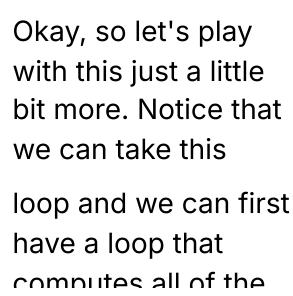
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what does it do. Every time it hits a new column, it already has been updated with the component orthogonal to the previous q's. It then goes and it one at the time computes a new coefficient in the appropriate row and subtracts out to the component corresponding to that coefficient.



$[A, R] := \text{GS}(A)$ (overwrites A with Q)

$$A \rightarrow \left(A_L \mid A_R \right), R \rightarrow \left(\begin{array}{c|c} R_{TL} & R_{TR} \\ \hline 0 & R_{BR} \end{array} \right)$$

A_L has 0 columns and R_{TL} is 0×0

while $n(A_L) < n(A)$

$$\left(A_L \mid A_R \right) \rightarrow \left(A_0 \mid a_1 \ A_2 \right), \left(\begin{array}{c|c} R_{TL} & R_{TR} \\ \hline 0 & R_{BR} \end{array} \right) \rightarrow \left(\begin{array}{c|cc} R_{00} & r_{01} & R_{02} \\ \hline 0 & \rho_{11} & r_{12}^T \\ 0 & 0 & R_{22} \end{array} \right)$$

CGS	MGs	MGs (alternative)
$r_{01} := A_0^H a_1$		
$a_1 := a_1 - A_0 r_{01}$	$[a_1, r_{01}] = \text{Proj}_{\perp \text{to } Q_{\text{MGs}}}(A_0, a_1)$	
$\rho_{11} := \ a_1\ _2$	$\rho_{11} := \ a_1\ _2$	$\rho_{11} := \ a_1\ _2$
$a_1 := a_1 / \rho_{11}$	$q_1 := a_1 / \rho_{11}$	$a_1 := a_1 / \rho_{11}$
		$r_{12}^T := a_1^H A_2$
		$A_2 := A_2 - a_1 r_{12}^T$

```

    ( [ A_L | A_R ] ← ( [ A_0  a_1 | A_2 ], ( [ 0 | R_{BR} ] ← ( [ 0  ρ_{11} | r_{i2} ]
endwhile

```

as

```
function [ Q, R ] = MGS_QR( A )
```

by completing the code in [Assignments/Week03/matlab/MGS_QR.m](#).
Input is an $m \times n$ matrix A . Output is the matrix Q and the upper triangular matrix R . You may want to use [Assignments/Week03/matlab/test_MGS_QR.m](#) to check your implementation.

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